ArmyForNER at KnowledgeGraphEval 2026: Multi-Stage Domain Adaptation and Self-Training Ensemble for Arabic Nested Named Entity Recognition

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Abstract

This paper presents the ArmyForNER system for Subtask 1 of the KnowledgeGraphEval 2026 shared task on Arabic named entity recognition (NER). We address nested, cross-domain Arabic NER with a multi-stage pipeline built on ARBERTv2: (1) domain-adaptive pre-training (DAPT) via masked language modeling on unlabeled *Konooz* text, (2) fine-tuning with a multi-head BIO tagging architecture on the Wojood nested NER corpus, and (3) self-training on pseudo-labels produced by an intermediate DAPT ensemble over the *Konooz* domains. Soft-voting three self-trained checkpoints obtains 74.85 micro-F1 and 65.20 macro-F1 on the official test set, against a 63.00 / 44.00 baseline. Because both adaptation stages consume the blind *Konooz* test sentences, our system is listed among those that used *Konooz* during training, and its score is not comparable to systems that used *Konooz* for evaluation only. We state this setting explicitly and find that self-training gives consistent gains across seeds, whereas the benefit of DAPT is configuration- and seed-dependent.

1 Introduction

Arabic is spoken natively by more than 400 million people, yet NLP systems for Arabic continue to lag behind those for resource-rich languages, largely due to the language's complex morphology and orthography (Darwish et al., 2021; Mashaabi et al., 2024). Arabic's root-and-pattern morphology packs information such as gender, number, person, aspect, mood, and clitics into a single word, and diacritics are typically omitted in written text, leaving a word with a dozen or more possible morphological analyses (Darwish et al., 2021; Qu et al., 2024). Arabic script also lacks capitalization, removing a signal that flat NER systems in other languages rely on heavily (Qu et al., 2024). These properties, together with the gap between Modern Standard Arabic and regional dialects, make Arabic NER particularly challenging.

Arabic NER has advanced substantially with resources such as the Wojood corpus (Jarrar et al., 2022), comprising approximately 448K tokens and 72K entity mentions across 21 (possibly nested) entity types. Models trained on Wojood, however, generalize poorly to new domains: the official KnowledgeGraphEval 2026 baselines show an unadapted AraBERTv02 model (Antoun et al., 2020) dropping from 0.63 overall micro-F1 to 0.43 on the Agriculture domain of the *Konooz* corpus (Hamad et al., 2025) (Table 1).

KnowledgeGraphEval 2026 Subtask 1 (AdaptNER) (Aljabari et al., 2026) was designed to assess exactly this generalization gap. Systems are trained on Wojood and then evaluated on ten previously unseen *Konooz* domains without any target-domain labels. Therefore, the task combines two challenges: domain shift and nested entity structure.

In this paper, we present a nested NER system built to be robust to this domain shift. Our system uses DAPT to reduce the domain gap, a multi-head Beginning-Inside-Outside (BIO) architecture for nested spans, and self-training with ensemble-derived pseudo-labels. A 3-model soft-vote ensemble over the self-trained checkpoints produces the final predictions (Figure 1). Our submission obtained 74.85 micro-F1 and 65.20 macro-F1 on the official test set. Because both the DAPT and the self-training stages consume the blind *Konooz* test sentences, our system falls outside the group of systems that used *Konooz* for evaluation only, and the comparison in Section §5 is framed accordingly. Our contributions are:

- A multi-head BIO architecture that predicts nested entity over a shared encoder.
- A three-stage adaptation pipeline: DAPT, fine-

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Konooz Domain	Baseline F_1
Economics	0.68
Art	0.67
History	0.67
Politics	0.67
Health	0.66
Law	0.66
Science	0.65
Finance	0.61
Sport	0.58
Agriculture	0.43
Overall (AraBERTv02)	**0.63**

Table 1: Official unadapted AraBERTv02 baseline micro-F1 per Konooz domain (Antoun et al., 2020; Aljabari et al., 2026).

tuning, and pseudo-label self-training, that adapts a Wojood-trained model to unseen Konooz domains without target-domain labels.

- An empirical comparison showing self-training yields consistent gains across seeds while DAPT's benefit is configuration-dependent (Section §5).

2 Background

The KnowledgeGraphEval 2026 shared task decomposes Arabic knowledge graph construction into cross-domain NER (Subtask 1: AdaptNER) and relation extraction between identified entities (Subtask 2) (Aljabari et al., 2026). This paper only addresses Subtask 1.

For Subtask 1, models are trained on the Wojood NER corpus (448K tokens, 72K entity mentions, 21 entity types) (Jarrar et al., 2022) and evaluated on the *Konooz* MSA corpus (Hamad et al., 2025): 1,645 sentences (≈50K tokens) drawn from ten specialized domains (Agriculture, Art, Economics, Finance, Health, History, Law, Politics, Science, and Sport), with no target-domain annotations available during development.

Task challenges. (i) *Nested named entities:* entities in Wojood and *Konooz* can overlap (e.g., the geopolitical entity "Cairo" nested within the organization "Cairo University"), requiring a multi-column sequence-labeling format across 21 annotation columns (Aljabari et al., 2026). Nested NER is commonly approached with layered taggers (Ju et al., 2018) or second-best span decoding (Shibuya and Hovy, 2020). Our multi-head formulation (Section §3.1) is a simpler alternative that maps directly onto this multi-column format. (ii) *Domain generalization:* sequence labellers trained on news-domain text degrade substantially on specialized domains without target-domain training data, as Table 1 shows.

3 System Overview

3.1 Base Model and Architecture

We use **UBC-NLP/ARBERTv2** (Abdul-Mageed et al., 2021) as our encoder backbone: an Arabic BERT model with 12 layers, 768 hidden dimensions, and 163M parameters. For nested NER, we adopt a multi-head BIO architecture. The shared encoder produces a contextualized token representation, and 21 independent linear classification heads, one per entity type in the Wojood annotation guidelines, each predict `O`, `B`, or `I`. Any token can therefore be labeled with several entity types at once. This is equivalent to solving 21 BIO sequence-labeling tasks in parallel over a shared encoder, and avoids enumerating candidate spans.

Training objective. The loss is the mean cross-entropy across all 21 heads, computed only over non-padded, first-subword positions:

$$\mathcal{L} = \frac{1}{21} \sum_{t=1}^{21} \mathrm{CE}(\mathbf{W}_t \, \mathbf{h}, y_t) \qquad (1)$$

where $\mathbf{h}$ is the encoder output, $\mathbf{W}_t \in \mathbb{R}^{3 \times H}$ is the head for entity type t, and y_t is the gold BIO label.

BIO repair. Because the heads decode independently per token, they can emit invalid tag sequences. At inference we therefore apply a deterministic repair pass to each head's tag sequence, scanning left to right: an `I` tag that is not preceded by a `B` or `I` tag of the same entity type is rewritten as `B`, opening a new span at that position. No other tags are modified.

3.2 Domain-Adaptive Pre-Training (DAPT)

To bridge the gap between the Wojood training domain and the *Konooz* evaluation domains, we apply DAPT (Gururangan et al., 2020): we continue masked language modeling (MLM) on unlabeled target-domain text before fine-tuning for NER. The DAPT corpus is the set of blind test sentences from all ten *Konooz* domains (Section §4.1), with a masking probability of 0.15.

We trained three DAPT variants to create diverse adapted encoders for ensembling. All three use a

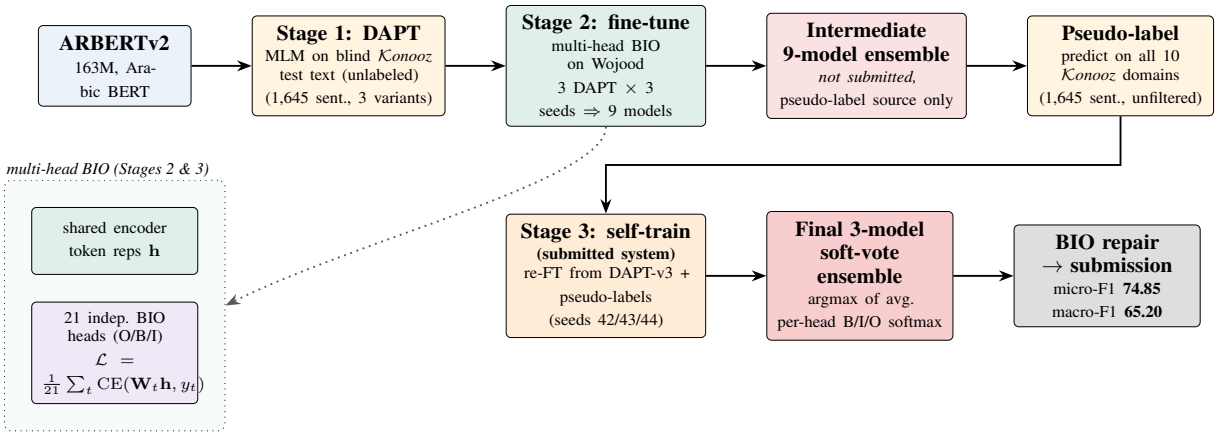


Figure 1: Our NER pipeline. DAPT adapts ARBERTv2 to the unlabeled *Konooz* text. The resulting encoders are fine-tuned with a multi-head BIO architecture on Wojood. Self-training then re-fine-tunes three new seeds from the DAPT-v3 checkpoint on Wojood plus pseudo-labels produced by the intermediate 9-model ensemble. Stage 1 and the pseudo-labeling step both consume the blind *Konooz* test sentences (without labels), making the pipeline transductive (§4.1).

learning rate of 5×10^{-5}; they differ as follows: DAPT-v1 and DAPT-v2 both run for 3 epochs with batch size 32 and differ only in random seed, while DAPT-v3 runs for 5 epochs with batch size 16. These settings were chosen to vary training length and batch size within our GPU budget rather than by a systematic search, which limits what the comparison in Section §5.3 can establish.

3.3 Fine-Tuning on Wojood

Each DAPT-adapted encoder is fine-tuned on the Wojood 1.1 nested NER corpus with learning rate 3×10^{-5}, batch size 16, weight decay 0.01, linear warmup over the first 10% of steps, gradient clipping at 1.0, maximum sequence length 256, and mixed-precision (FP16) training. We train for up to 10 epochs and keep the checkpoint with the best micro-F1 on the Wojood validation set. Three seeds (42, 43, 44) per DAPT variant give nine fine-tuned models, which are combined into an intermediate soft-vote ensemble used solely to generate pseudo-labels. This intermediate ensemble is not part of the submitted system.

3.4 Self-Training with Ensemble Pseudo-Labels

We run the intermediate 9-model ensemble on the 1,645 unlabeled *Konooz* sentences and keep all of its predictions as pseudo-labels, with no confidence threshold or agreement filtering. These sentences are appended to the Wojood training set (23,124 + 1,645 = 24,769 sentences), and we fine-tune three new seeds (42, 43, 44) starting from the DAPT-v3 checkpoint.

We note that DAPT-v3 was not the validation-preferred variant: DAPT-v2 was the most consistent on Wojood validation (Section §5.3). The DAPT-v3 checkpoint was selected before the per-variant validation comparison was complete, and we did not re-run self-training from DAPT-v2 within the shared-task timeline. Whether initializing from DAPT-v2 would have performed better therefore remains untested.

3.5 Soft-Vote Ensemble

Our final submission combines the three self-trained checkpoints by soft voting: for each token position and entity-type head, we average the three-way softmax distributions and select the highest-probability BIO tag. BIO repair is then applied to the ensemble output.

4 Experimental Setup

Data. We train on the Wojood 1.1 nested NER corpus (Jarrar et al., 2022): 23,124 training sentences (≈391K tokens) and 3,303 validation sentences (≈58K tokens) annotated with 21 entity types. The official test set is drawn from the *Konooz* corpus (Hamad et al., 2025), covering ten Arabic domains.

Hardware. All experiments were run on a single NVIDIA RTX 3070 Ti GPU (8 GB VRAM).

Implementation. Our code is built on HuggingFace Transformers (version 4.48.1) and PyTorch (version 2.2.2).

	Team	P	R	Mi-F1	Ma-F1
Konooz used for evaluation only					
1	NAMAA	83.83	69.44	75.96	66.00
2	Wasl	80.60	69.82	74.83	63.16
3	McMasterNLP	75.00	74.00	74.38	65.00
4	PTUK-NLP	71.84	66.59	69.11	60.70
Konooz used during training or adaptation					
5	**ArmyForNER**	**82.11**	**68.77**	**74.85**	**65.20**
6	AdaptLinker	75.48	71.21	73.29	63.37
7	YafaNER	73.25	69.52	71.33	60.91
8	ANLP-UniSo	81.43	63.04	71.06	61.01
–	Baseline	63.87	62.30	63.00	44.00

Table 2: Final official Subtask 1 results. The first group used *Konooz* only for evaluation; the second, including our system, used *Konooz* data during training or adaptation.

4.1 Use of Target-Domain Data

Both adaptation stages of our pipeline consume the blind *Konooz* test sentences: DAPT performs MLM on their raw text, and self-training adds pseudo-labels for those same sentences to the training set. No *Konooz* gold labels were used at any point, and no labeled target-domain data was available to us. Our setting is therefore transductive: the evaluation sentences themselves, without their labels, are part of both adaptation stages.

In the official results, this places our system among those that used *Konooz* data during training rather than among the systems that were used *Konooz* for evaluation only. Scores across the two groups are not directly comparable, and we do not claim a rank relative to the systems in the first group.

5 Results and Analysis

5.1 Official Test Results

Table 2 reports the final official results. Our system obtains 74.85 micro-F1 and 65.20 macro-F1, improving over the official baseline by 11.85 and 21.20 points respectively. Precision (82.11) exceeds recall (68.77) by more than 13 points, so the residual error is dominated by entity mentions the system misses rather than by spurious predictions, which is the expected failure mode when a news-trained tagger meets specialized domain terminology.

The organizers report only aggregate scores, so we cannot give a per-domain *Konooz* breakdown and cannot tell whether the baseline's sharp drop on Agriculture (Table 1) was recovered.

5.2 Development Results

Table 3 reports micro-F1 on the Wojood validation set for each variant. These are source-domain scores: they measure the model on Wojood, not on the *Konooz* target domains, and §5.5 shows that they do not predict target-domain performance. We report them because they are the only labeled signal we had during development.

Model	Dev F1	Δ
ARBERTv2 (no DAPT, seed 42)	0.8297	–
DAPT-v1 + FT (seed 42)	0.8434	+1.37
DAPT-v1 + FT (seed 43)	0.8267	−0.30
DAPT-v1 + FT (seed 44)	0.8273	−0.24
DAPT-v2 + FT (seed 42)	0.8374	+0.77
DAPT-v2 + FT (seed 43)	0.8378	+0.81
DAPT-v2 + FT (seed 44)	0.8326	+0.29
DAPT-v3 + FT (seed 42)	0.8285	−0.12
DAPT-v3 + FT (seed 43)	0.8313	+0.16
DAPT-v3 + FT (seed 44)	0.8277	−0.20
Self-trained v3 (seed 42)	0.8335	+0.50
Self-trained v3 (seed 43)	0.8377	+0.64
Self-trained v3 (seed 44)	0.8444	+1.67

Table 3: Micro-F1 on the *Wojood validation set*. Δ is in F1 points, against the no-DAPT baseline for the DAPT rows and against the corresponding DAPT-v3 seed for the self-trained rows.

5.3 Effect of DAPT

DAPT does not help uniformly. DAPT-v1 gains 1.37 points for seed 42 but loses ground for seeds 43 and 44 (-0.30, -0.24). DAPT-v2 improves all three seeds ($+0.77$, $+0.81$, $+0.29$). DAPT-v3 stays near the baseline, ranging from -0.20 to $+0.16$. On Wojood validation, then, DAPT-v2 was the most consistently beneficial variant, and no single DAPT effect can be read off these numbers. Since these are source-domain scores, they also do not establish what DAPT contributed on *Konooz*. Isolating that would require target-domain labels or an official per-configuration submission, neither of which was available.

5.4 Effect of Self-Training

Self-training on top of DAPT-v3 improves every seed relative to its non-self-trained counterpart: $+0.50$, $+0.64$, and $+1.67$ points. Unlike the DAPT gains, this direction is consistent across all three seeds we ran, which is why we adopted self-training for the final system. The same caveat applies: the gain is measured on Wojood, while the pseudo-labels come from *Konooz*.

5.5 Source-Target Gap

Our Wojood validation scores cluster around 0.83-0.84, while our official *Konooz* micro-F1 is 0.7485, a drop of roughly 8-9 points. This is the cross-domain degradation the task is designed to expose, and it persists despite both adaptation stages: DAPT and self-training narrow the gap relative to the 0.63 baseline but do not close it.

The two numbers are not directly comparable, since they come from different corpora with different domain and label distributions, and the official metric is span-level over *Konooz*. The practical consequence for system building is that ranking variants by Wojood validation F1 is unreliable: the differences in Table 3 are mostly well under one point and of the same order as seed noise, and none of them was validated against target-domain performance.

5.6 Error Analysis

Manual inspection of system errors on the Wojood validation set reveals three main error categories. Because *Konooz* gold labels are not available to us, this analysis is conducted entirely on the source domain and is at best indirect evidence about target-domain behavior.

Boundary errors. A substantial fraction of errors are off-by-one span boundaries, where the entity type is correct but the span includes or excludes one token.

Nested entity confusion. The multi-head design permits nested predictions, but the model sometimes fails to predict the outer span when the inner entity is more salient.

Rare and domain-specific entities. Specialized terminology is under-represented in Wojood's largely news-domain text. The precision–recall asymmetry in the official results (82.11 vs. 68.77) is consistent with such mentions going undetected on *Konooz*, though we cannot confirm this without target-domain labels.

6 Limitations

Our system has several limitations. First, and most importantly, both adaptation stages use the blind *Konooz* test sentences (§4.1). The resulting system is transductive: it cannot be applied to a new domain without first collecting that domain's text and re-running DAPT and self-training, and its score is not comparable to systems that never touch the evaluation text. We did not measure how much of our gain over the baseline would survive if adaptation were restricted to a separate unlabeled pool, such as the provided *Konooz*-dev text.

Second, every ablation number we report is Wojood validation F1, a source-domain measurement, while DAPT and self-training are both aimed at the target domain. We therefore cannot attribute our official score to any individual stage, and we did not obtain official per-configuration scores that would allow us to.

Third, the pseudo-labels were used without confidence or agreement filtering, so systematic errors of the intermediate ensemble on specialized terminology may have been reinforced rather than corrected; we ran no filtered-versus-unfiltered comparison. Fourth, the self-training checkpoint was initialized from DAPT-v3 although DAPT-v2 was the validation-preferred variant (§3.4), and this choice was never tested. Fifth, the multi-head loss weights all 21 entity types equally regardless of frequency, which disadvantages rare types and plausibly contributes to the low recall in the official results. Sixth, producing the pseudo-labels required training and running a 9-model intermediate ensemble, a compute cost far larger than the submitted 3-model ensemble. Finally, each configuration was trained with only three seeds and the differences between several variants are smaller than the observed seed spread, so individual comparisons in Table 3 should be read as indicative rather than conclusive.

7 Conclusion

We presented a multi-stage system for Arabic nested NER combining domain-adaptive pretraining, multi-head BIO fine-tuning, and self-training with ensemble-derived pseudo-labels, submitted as a 3-model soft-vote ensemble. The system obtained 74.85 micro-F1 and 65.20 macro-F1 on the *Konooz* evaluation benchmark, well above the 63.00 / 44.00 baseline, in a transductive setting that uses the blind test sentences for both adaptation stages. Self-training gave a more consistent gain across seeds than DAPT, and Wojood validation performance did not predict *Konooz* test performance. The main open questions our experiments leave are whether confidence-filtered pseudo-labels would help, whether initializing self-training from the validation-preferred DAPT variant would change the outcome, and how much of the gain

survives when adaptation is restricted to a separate unlabeled target-domain pool.

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